Outline Draft

Loveless, Nix, Stokes, Abernathy's

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1 Introduction

This serves as an outline for equations of our system of equations for research. Eventually the paper outline may be formed from this document...

2 Equations

Potential systems consist of various mechanisms, but the main differences come from whether or not we contain a Holling-Type III term in our A_β compartment. The other difference comes from whether or not we include a logistic growth term in each compartment [2], [1]. The Equations including the Holling-Type III term is given by

$$\begin{aligned}
\frac{dA_\beta}{dt} &= \lambda_{A_\beta} A_\beta (K_{A_\beta} - A_\beta) + \lambda_{A_\beta Ca} \frac{Ca^2}{Ca^2 + \sigma^2} - (1 - \alpha_{A_\beta}) \delta_{A_\beta} A_\beta && (1) \\
\frac{dCa}{dt} &= \lambda_{Ca} Ca (K_{Ca} - Ca) + \lambda_{Ca A_\beta} - (1 - \alpha_{Ca}) \delta_{Ca} Ca && (2) \\
\frac{d\tau}{dt} &= \lambda_\tau \tau (K_\tau - \tau) + \lambda_{\tau A_\beta} A_\beta + \lambda_{\tau Ca} - (1 - \alpha_\tau) \delta_\tau \tau && (3) \\
\frac{dN}{dt} &= \lambda_N N (K_N - N) + \lambda_{N\tau} \tau && (4) \\
\frac{dC}{dt} &= \lambda_C C (K_C - C) + \lambda_{CN} NR + \lambda_{C\epsilon} \epsilon. && (5)
\end{aligned}$$

A simpler model can be defined, excluding the Holling-Type III term within dA_β/dt.

$$\begin{aligned}
\frac{dA_\beta}{dt} &= a_1 - k_1 A_\beta + \lambda_{A_\beta Ca} - (1 - \alpha_{A_\beta}) \delta_{A_\beta} A_\beta && (6) \\
\frac{dCa}{dt} &= a_2 - k_2 Ca + \lambda_{Ca A_\beta} - (1 - \alpha_{Ca}) \delta_{Ca} Ca && (7) \\
\frac{d\tau}{dt} &= a_3 - k_3 \tau + \lambda_{\tau A_\beta} A_\beta + \lambda_{\tau Ca} - \underbrace{(1 - \alpha_\tau) \delta_\tau \tau}_{\text{off the dome}} && (8) \\
\frac{dN}{dt} &= a_4 - k_4 N + \lambda_{N\tau} \tau && (9) \\
\frac{dC}{dt} &= \underbrace{a_5 - k_5 C}_{\text{Pal}} + \underbrace{\lambda_{CN} NR + \lambda_{C\epsilon} \epsilon}_{\text{Petrella}}. && (10)
\end{aligned}$$

In each system, a_n, k_n is the growth and wash out, λ_{ab} is the growth rate for a with respect to b, $ab \neq 0$. α_n is the inefficiency of some drug for n, $0 < \alpha_n < 1$. δ_n is the strength of some drug, $0 < \delta_n < 1$.

The proposed **MODEL TO END ALL MODELS**:

$$\begin{aligned}
\frac{dA_\beta}{dt} &= a_1 - r_1 A_\beta + a_2 \frac{Ca^2}{Ca^2 + \sigma^2} && (11)\\
\frac{dCa}{dt} &= b_1 - r_2 Ca + b_2 A_\beta && (12)\\
\frac{d\tau}{dt} &= c_1 - r_3 \tau + c_2 A_\beta + c_3 Ca && (13)\\
\frac{dN}{dt} &= d_1 - k_4 N + d_2 \tau && (14)\\
\frac{dC}{dt} &= e_1 - k_5 C + e_2 NR. && (15)
\end{aligned}$$

where $r_n = k_n + (1 - \alpha_n)\delta_n$.

References

[1] Swadesh Pal and Roderick Melnik. "Nonequilibrium landscape of amyloid-beta and calcium ions in application to Alzheimer's disease". In: *Physical Review E* 111.1 (2025), p. 014418.

[2] Jeffrey R Petrella et al. "Computational causal modeling of the dynamic biomarker cascade in Alzheimer's disease". In: *Computational and mathematical methods in medicine* 2019.1 (2019), p. 6216530.